

Lemma 8.1.6. *Suppose $\ell \neq 2$ and $(\ell, n!) = 1$ if we are in case O1. If $\phi \in \Phi^{\text{ss}}(G^*, \overline{\mathbb{F}}_\ell)$ is a generic semi-simple toral L -parameter, and $\mu \in X_\bullet(G^*)$ is a dominant cocharacter, then ϕ is μ -regular.*

Similarly, if $\phi^\sharp \in \Phi^{\text{ss}}(G^\sharp, \overline{\mathbb{F}}_\ell)$ is a generic semi-simple toral L -parameter, and $\mu^\sharp \in X_\bullet(G^\sharp)$ is a dominant cocharacter such that $(\ell, n!) = 1$ if we are in case O1, then $\phi^\sharp$ is $\mu^\sharp$ -regular.

In particular, $\mu^\sharp$ can be chosen to be not fixed by any non-trivial element of $W_{G^\sharp}$.

Proof. By base change [HL24, Lemma 4.22] and the isomorphism Equation (8.1), it suffices to prove for the general linear group $\text{GL}(n)$, the split special orthogonal group $\text{SO}(d(G))$ and the split general spin group $\text{GSpin}(d(G))$. For general linear groups (i.e., in case U) this is proved in [HL24, Lemma 4.22], and the argument also works for $G \in \{\text{SO}(d(G)), \text{GSpin}(d(G))\}$:

The standard representation $\widehat{\text{Std}}$ of $\text{Sp}(2n(G^*))$ or $\text{SO}(2n(G^*))$ has weights given by Weyl orbits of ω_1 . The standard representation extends to a standard representation of $\text{GSp}(2n(G^*))$ or $\text{GSO}(2n(G^*))$, and we write the highest weight of the standard representation of $\text{GSp}(2n(G^*))$ or $\text{GSO}(2n(G^*))$ as $\omega_1^\sharp$. Then it is clear that difference of the weights appearing in $\widehat{\text{Std}}$ are coroots of G , thus ϕ is strongly μ -regular and $\phi^\sharp$ is strongly $\mu^\sharp$ -regular, by definition of genericity. Thus, they are also μ -regular (resp. $\mu^\sharp$ -regular) by the proof of [Ham24, Theorem 10.10].

For other cocharacters, we first recall that with $\overline{\mathbb{Q}}_\ell$ -coefficients, the highest weight tilting module $\mathcal{T}_{\omega_i}$ associated to $\omega_i \in X_\bullet(\text{Sp}(2n(G^*)))$ is realized on the space of harmonic elements in $\wedge^i(\widehat{\text{Std}})$ as defined in [GW09, §5.5.2], and it extends to a standard representation of $\text{GSp}(2n(G^*))$ with highest weight denoted by $\omega_i^\sharp$, and the same is true with $\overline{\mathbb{F}}_\ell$ -coefficients, where each highest weight tilting module associated to $\omega_i^\sharp$ of $\text{GSp}(2n(G^*))$ is a direct sum of $\wedge^i(\widehat{\text{Std}})$, by our assumption on ℓ , cf. [Jan03, Page 286-287] and [Ham24, §10.1, Appendix B.2]. Similarly, with $\overline{\mathbb{Q}}_\ell$ -coefficients, $\wedge^i(\widehat{\text{Std}})$ is isomorphic to the highest weight tilting module $\mathcal{T}_{\omega_i}$ associated to $\omega_i \in X_\bullet(\text{SO}(2n(G^*)))$ for $1 \leq i \leq n(G^*) - 2$, and $\wedge^{n(G^*)-1}(\widehat{\text{Std}})$ is isomorphic to the highest weight tilting module $\mathcal{T}_{\omega_{n(G^*)-1} + \omega_{n(G^*)}}$ associated to $\omega_{n(G^*)-1} + \omega_{n(G^*)} \in X_\bullet(\text{SO}(2n(G^*)))$. On the other hand, $\wedge^{n(G^*)-1}(\widehat{\text{Std}})$ is isomorphic to the direct sum of highest weight tilting modules $\mathcal{T}_{2\omega_{n(G^*)-1}}$ and $\mathcal{T}_{2\omega_{n(G^*)}}$, cf. [GW09, Theorem 5.5.13], and these extends to a standard representation of $\text{GSp}(2n(G^*))$ with highest weight denoted by

$$\omega_1^\sharp, \dots, \omega_{n-2}^\sharp, \omega_{n-1}^\sharp + \omega_{n-2}^\sharp, 2\omega_{n-1}^\sharp, 2\omega_{n-2}^\sharp,$$

respectively. The same is true with $\overline{\mathbb{F}}_\ell$ -coefficients, where each highest weight tilting module associated to the above cocharacters of $\text{GSp}(2n(G^*))$ is a direct sum of $\wedge^i(\widehat{\text{Std}})$, by our assumption on ℓ , cf. [Jan03, Page 286-287] and [Ham24, §10.1, Appendix B.2].

Now all these highest weight tilting modules of $G^\sharp$ with fundamental weights $\mu^\sharp$ appear as direct summand of tensor products of $\widehat{\text{Std}}$, so $\phi^\sharp$ is $\mu^\sharp$ -regular by [Ham24, Proposition 10.12].

Finally, we can choose $\mu^\sharp$ appropriately such that under the isomorphism Equation (8.1)

$$G_{\overline{\mathbb{Q}}_p}^\sharp \cong \begin{cases} \text{GL}(1)_{\overline{\mathbb{Q}}_p} \times \prod_{v \in \text{Hom}(K, \overline{\mathbb{Q}}_p)} \text{GSpin}(V^* \otimes_{K,v} \overline{\mathbb{Q}}_p) & \text{in case O} \\ \text{GL}(1)_{\overline{\mathbb{Q}}_p} \times \prod_{v \in \text{Hom}(K, \overline{\mathbb{Q}}_p)} \text{GU}(V^* \otimes_{K_1,v} \overline{\mathbb{Q}}_p) & \text{in case U} \end{cases},$$

it is of the form $(0, \mu^{\sharp'}, \dots, \mu^{\sharp'})$ (i.e., trivial on the $\text{GL}(1)$ factor and identical on the other factors), where $\mu^{\sharp'}$ is not fixed by any non-trivial Weyl group element. Then $\mu^\sharp$ is not fixed by any non-trivial Weyl group element. $\square$

8.2. Perverse t -exactness and vanishing results. In this subsection, we prove a perverse t -exactness result for Hecke operators, and deduce a vanishing result for cohomology of Shimura varieties with torsion coefficients. We use notations related to Bun_G defined in Section 3.1.

For any reductive group G over a non-Archimedean local field K of characteristic 0 and any open substack $U \subset \text{Bun}_G$, there exists a perverse t -structure on $\text{D}_{\text{lis}}(\text{Bun}_G, \overline{\mathbb{F}}_\ell)$ defined as follows [HL24, Definition 4.11]: For each $\mathfrak{b} \in B(G)$, denote $d_{\mathfrak{b}} := \langle 2\rho_G, \nu_{\mathfrak{b}} \rangle$ where $\nu_{\mathfrak{b}}$ is the slope homomorphism of $\mathfrak{b}$, then an object A is contained in ${}^p\text{D}^{\leq 0}(U, \overline{\mathbb{F}}_\ell)$ if $i_{\mathfrak{b}}^* A \in \text{D}^{\leq d_{\mathfrak{b}}}(\mathbb{G}_{\mathfrak{b}}, \Lambda)$, and A is contained in ${}^p\text{D}^{\geq 0}(U, \overline{\mathbb{F}}_\ell)$ if $i_{\mathfrak{b}}^! A \in \text{D}^{\geq d_{\mathfrak{b}}}(\mathbb{G}_{\mathfrak{b}}, \Lambda)$. Here we recall that $i_{\mathfrak{b}}$ is the inclusion $\text{Bun}_{\mathfrak{b}}^{\mathfrak{b}} \subset \text{Bun}_G$.

We also need the notion of universally locally acyclic (ULA) objects [FS24, Definition IV.2.31]: The full subcategory $\text{D}^{\text{ULA}}(\text{Bun}_G, \overline{\mathbb{F}}_\ell) \subset \text{D}_{\text{lis}}(\text{Bun}_G, \overline{\mathbb{F}}_\ell)$ consists of objects A such that $i_{\mathfrak{b}}^* A \in \text{D}^{\text{adm}}(\mathbb{G}_{\mathfrak{b}}, \overline{\mathbb{F}}_\ell)$ for each $\mathfrak{b} \in B(G)$.

We import Setup 7.2.2. In particular, F is a totally real number field unramified at a prime p , $\mathbf{G}$ is a special orthogonal or unitary group over F with $G^* = \mathbf{G} \otimes_F (F \otimes \mathbb{Q}_p)$, and $G^\sharp$ is a central extension of $\mathbf{G}$ with $G^\sharp$. Then we have the following local result on the perverse t -exactness of Hecke operators, which generalizes [HL24, Corollary 4.24] to special orthogonal groups and unitary groups: